Piyush Chaudhary

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SUMMARY

Final-year B.Tech Computer Engineering student building document-processing, cloud and mobile software. Three internships: State Bank of India (two) and Mazagon Dock Shipbuilders.

EDUCATION

Bachelor of Technology, Computer Engineering, K.J. Somaiya College of Engineering — Final year
Coursework: Natural Language Processing, Cloud Computing, AR/VR, Network and Information Security

EXPERIENCE

Summer Intern, State Bank of India, Analytics Department — May 2026 - Jul 2026

- Project: image to structured dataset conversion and predictive model building. Built a production pipeline that extracts structured tables from scanned telecom audit PDFs into Excel, using a hybrid of computer vision (PyMuPDF, OpenCV) and a vision-language model, chosen after comparing three architectures.
- Added a fast pre-scan for key header fields and the conclusion-table pages, and switched to lazy page generators after eager rendering exhausted memory.
- Fixed a bug that turned an 11-column table into 24 phantom columns, added a guard against implausible model output, tested on synthetic noisy scans, and wrote a 16-page technical report.

Industrial Training Intern, Mazagon Dock Shipbuilders Limited, CIT Department — Jun 2025
Summer Intern, State Bank of India, Business Intelligence Department — Jun 2024

- Project: Dashboard Access Management Tool. Completed the project and submitted a project report.

PROJECTS

namestorage: Self-Hosted Cloud Drive (Python, WebRTC, WebSockets, STUN/ICE, Tailscale Funnel, OIDC, systemd, Linux, pytest) — github.com/Piyushchy/namestorage

- Private drive on my own Ubuntu laptop behind CGNAT: the agent makes only outbound connections (no port forwarding), and the browser connects peer-to-peer over an encrypted WebRTC data channel, so no server stores or relays files.
- NAT traversal with ICE and Google's public STUN server. Published the Python WebSocket signaling server as a public wss:// endpoint with Tailscale Funnel (no VPS, static IP or open port); it relays only SDP setup messages.
- Security: server-side verification of Google OIDC ID tokens (signature, issuer, audience, expiry) against an email allow-list on both server and agent; origin check; rate limiting; single-use HMAC tickets (120 s, replay-rejected) bound to the DTLS fingerprint; path-traversal confinement; systemd hardening.
- One-command installer and management CLI. 79 automated tests (pytest), including a real WebRTC session.

The Tangled Path: Choice-Based Anti-Drug Game (HTML, CSS, JavaScript) — sudo-choices.vercel.app

- 1st runner up at a hackathon (team of four; I was team leader and front-end developer). A story game across four life stages (ages 6 to 70) where each choice leads to a good or bad ending. Deployed on Vercel.

MouseShare (Python, Kotlin, Android)

- Built because I could not find an existing tool that shares a Windows PC's mouse and keyboard with an Android phone: a Python desktop app captures input and a Kotlin app receives it. Comparing transports (HID, WiFi) to cut input lag.

GIF Crop and Overlay Tool (JavaScript, HTML, Canvas) — piyushchy.github.io/gif-tool

- Built because no tool cropped a GIF to a freehand shape or overlaid one GIF on another with frame-level control. Wrote the GIF decoder and encoder from scratch (LZW, median-cut palette); runs in the browser.

Cloud-Native App on AWS (Docker, AWS ECR, ECS Fargate, ALB, RDS MySQL, CodePipeline, SNS)

- Containerized a Node.js app and deployed it on ECS Fargate behind an Application Load Balancer, with ECR, RDS MySQL (reached through a bastion host), a CodePipeline CI/CD flow and an SNS alarm.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, HTML, CSS
Web: React.js, HTML5, CSS3, WebRTC, WebSockets
Computer vision and ML: OpenCV, PyMuPDF, EasyOCR, Table Transformer, Docling, vision-language models, PyTorch, RNN, LSTM
Cloud: Docker, AWS (ECR, ECS Fargate, ALB, RDS MySQL, EC2, CodePipeline, SNS)
Tools and systems: Linux, Command Line (CLI), SSH, Bash, systemd, Tailscale, Git, GitHub, pytest, Google Colab, openpyxl